Unit: 6 Unit 7: Factoring **Name:** ________________
Topic: Expanding Special Patterns Perfect Square Trinomial **Date:** ___________

Advanced (Habanero)

Q1. What is the expansion of $(x+y)^2$?
(A) $x^2+2xy+y^2$
(B) $x^2-2xy+y^2$
(C) x^2+xy+y^2
(D) x^2-xy+y^2
(E) x^2+y^2

Q2. Which of the following is a perfect square trinomial?
(A) x^2+4x+4
(B) x^2-4x-4
(C) x^2+4x-4
(D) x^2-4x+4
(E) x^2+4x+2

Q3. What is the factored form of $x^2+10x+25$?
(A) $(x+5)^2$
(B) $(x-5)^2$
(C) $(x+5)(x-5)$
(D) $(x+2)(x+8)$
(E) $(x-2)(x-8)$

Q4. Which of the following is the expansion of $(x-y)^2$?
(A) $x^2+2xy+y^2$
(B) $x^2-2xy+y^2$
(C) x^2+xy+y^2
(D) x^2-xy+y^2
(E) x^2+y^2

Q5. What is the factored form of $x^2-12x+36$?
(A) $(x+6)^2$
(B) $(x-6)^2$
(C) $(x+6)(x-6)$
(D) $(x+2)(x-8)$
(E) $(x-2)(x+8)$

Q6. Which of the following is a perfect square trinomial?
(A) x^2+6x+9
(B) x^2-6x-9
(C) x^2+6x-9
(D) x^2-6x+9
(E) x^2+6x+2

Q7. What is the expansion of $(x+y)^2$ using the formula $a^2+2ab+b^2$?
(A) $x^2+2xy+y^2$
(B) $x^2-2xy+y^2$
(C) x^2+xy+y^2
(D) x^2-xy+y^2
(E) x^2+y^2

Q8. Which of the following is the factored form of $x^2+14x+49$?
(A) $(x+7)^2$
(B) $(x-7)^2$
(C) $(x+7)(x-7)$
(D) $(x+2)(x+12)$
(E) $(x-2)(x-12)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^2+20x+100$. Show all steps and explain your reasoning.

Q10. Expand the expression $(x-3)^2$ using the formula $a^2-2ab+b^2$. Show all steps and explain your reasoning.

Chef's Tips

- Tip 1: Ensure students understand the difference between expanding and factoring.
- Tip 2: Encourage students to use the formulas for special patterns to expand binomials.
- Tip 3: Remind students to check their work by multiplying the factored form to verify it equals the original expression.